When Does Generative AI Preserve or Undermine Signaling? Asymmetric Cost Reductions in Hiring Markets

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Abstract

In hiring markets, employers evaluate candidates under conditions in which their underlying abilities are only imperfectly observable. To reduce this uncertainty, they often rely on written signals such as CVs, cover letters, and other forms of self-presentation. Traditionally, signaling theory has explained the credibility of these written signals; whereas for low-quality workers producing a signal carries high cost, while for high-quality workers producing the same signal costs less. Generative AI may fundamentally alter the credibility of written signals by reducing the cost of producing polished communication. This cost reduction may advantage candidates non-uniformly. This thesis develops a signaling model to examine how generative AI changes the credibility of written application materials in hiring, with particular attention to differences between the high- and low-type candidates. The results show that AI weakens signal credibility when it functions as an equalizer that benefits low-type candidates disproportionately, but can strengthen it when AI instead augments high-type candidates' existing advantage. The thesis concludes that generative AI's effect on hiring signals is therefore conditional on its distributional impact across candidate types, rather than uniformly positive or negative.

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1 Introduction

Employers today increasingly receive job applications that have been polished, or entirely drafted, with the help of generative AI (Lievens & Dunlop, 2025). A well-written CV or cover letter has traditionally been viewed as a credible signal of the candidate's underlying ability and effort. When generative AI can help produce that same polished output for a much lower effort cost, employers fear a new and practical problem: they can no longer be certain that the quality of a written application still reflects the quality of the candidate submitting it (Robie et al., 2026).

This problem is a specific instance of a more general and long-studied phenomenon: information asymmetry between two parties to a transaction, where one side cannot directly verify a characteristic that is important to the other. Akerlof (1970) first formalized this concept, laying the foundation for a large body of research on markets characterized by asymmetric information. Building on this foundation, Spence (1973) extended the analysis to the labor market through signaling theory. He argued that observable actions can credibly reveal otherwise unobservable characteristics, provided they are systematically less costly for high-quality individuals to produce than for low-quality individuals.

Written application materials plausibly function as a signal in the sense of Spence's (1973) signaling theory. Producing a polished CV or cover letter typically requires time, effort, and writing ability, all of which are expected to differ across candidates (Connelly et al., 2011). As a result, differences in application quality may provide employers with information about underlying candidate quality. This thesis focuses on written application materials, because generative AI directly affects the cost of producing them, thereby potentially altering their signaling value. How generative AI changes these relative production costs is theoretically ambiguous. If AI reduces the cost of producing written application materials relatively more for lower-quality candidates, these materials may become less informative about candidate quality. Conversely, if higher-quality candidates benefit relatively more from AI, the informational value may be preserved or even strengthened. Which of these outcomes emerges depends on how generative AI changes the relative costs of producing written application materials, a question that remains theoretically unresolved.

This ambiguity motivates the central research question of this thesis: under what conditions does generative AI weaken the credibility of written application materials as a signal of candidate quality in hiring, and under what conditions does it preserve that credibility? To answer this question, this thesis develops a signaling model grounded in the incentive-compatibility logic of Spence (1973). Here, generative AI is introduced as a parameter that reduces the cost of producing a written signal, and in which this reduction can differ across candidate types. The model is deliberately kept simple, using binary candidate types and a binary signal. This enables the conceptually defined equalizer and augmentor scenarios in Lievens and Dunlop (2025) to be formalized as explicit conditions on the model's parameters. Each resulting scenario is evaluated against a common benchmark: whether a separating equilibrium holds, meaning written signals remain informative about candidate quality.

The remainder of the thesis proceeds as follows. Section 2 reviews the literature on information asymmetry, written application materials as hiring signals, and the productivity effects of generative AI. Section 3 forms the theoretical bridge from this literature to the modelling choices. Section 4 presents the formal model, moving from a pre-AI baseline to the possible scenarios following the introduction of generative AI. Section 5 discusses what these results imply for the credibility of written application materials in hiring.

Generative AI tools were used during the development of this thesis for language editing, feedback on modelling choices, and assistance in reviewing the presentation of the formal model. The author remained responsible for the theoretical assumptions, mathematical derivations, interpretation, references, and final content of the thesis. Documentation of substantial generative AI interactions is available through the associated OSF repository.

2 Literature review

2.1 Information asymmetry and signaling in hiring

Hiring is a classical case of information asymmetry: employers evaluate candidates under conditions in which their underlying abilities are only imperfectly observable. Left unaddressed, this kind of asymmetry can produce adverse selection. Akerlof (1970) demonstrated this in the market for used cars. Buyers cannot observe true quality, therefore high-quality sellers cannot obtain a price that reflects their value, while low-quality sellers free-ride on the average price buyers are willing to pay and potentially drive good cars out of the market. Spence (1973) introduced the mechanism by which the problem of adverse selection can be mitigated: costly signaling. Several items in the hiring context, such as education credentials, work experience, and written materials, might plausibly function as signals whose costs differ systematically across candidate types. Such signals may help employers form inferences about underlying ability, despite the information asymmetry inherent in hiring. If an observable action is differentially costly across types: cheaper for a high-productivity individual to produce than for a low-productivity individual, that action can credibly reveal an individual's underlying abilities. It is this differential-cost property, rather than the content of the signal itself, that provides the explanatory power in Spence's framework, and it is the property this thesis relies on throughout.

Signaling theory's explanatory power depends on the differential-cost property actually holding, rather than merely being assumed. Bergh et al. (2014) warn that much empirical work using signaling theory does not verify whether the separating equilibrium the theory presumes is in fact present. This risks treating the theory as a label rather than testing its underlying mechanism. This thesis takes that caution seriously. Section 4 derives, for each scenario considered, the explicit conditions under which a separating equilibrium exists.

2.2 Written application materials as hiring signals

Whether the signaling mechanism described above applies to hiring in practice, depends on whether the observable actions available to candidates satisfy the differential-cost property. Written application materials, such as CVs, cover letters, and written communication more broadly, are observable outputs that can possibly serve as a signal. Producing a polished and well-argued application plausibly draws on the same underlying reasoning and writing ability that predicts productivity on the job. A high-productivity candidate can generate such quality with comparatively little effort, whereas a low-productivity candidate must invest disproportionately more to produce the same quality. This is precisely the single-crossing property Spence's model requires.

Despite the central role of written application materials in the signaling process, existing research on AI in hiring has focused primarily on the employer's side of the selection process. Comparatively little attention has been paid to what happens once applicants themselves gain access to generative AI. In particular, the implications for the credibility of written application materials remain underexplored. If applicants can use generative AI to produce polished written materials at substantially lower cost, the informational value of these materials may change. This is the gap this thesis addresses. Applicant-side use of generative AI in job applications has grown rapidly since the technology became widely available. This trend is evidenced by studies on AI in personnel selection (Lievens & Dunlop, 2025; Robie et al., 2026). The observed productivity gains associated with generative AI assisted writing help explain this development (Noy & Zhang, 2023; Brynjolfsson et al., 2023). Additional contextual evidence comes from broader analyses of AI-assisted knowledge work (Berger et al., 2024; Dell'Acqua et al., 2026).

2.3 Generative AI and the production of written signals

To assess how generative AI affects the credibility of written hiring signals, it is first necessary to understand its broader effects on the cost and quality of producing written materials. The evidence reviewed here concerns professional writing and knowledge work broadly, rather than CVs, cover letters, or hiring outcomes. These studies do not speak directly to how AI affects the credibility of hiring signals. What these studies demonstrate is that generative AI systematically affects both the cost and the quality of written production. This thesis treats that general pattern explicitly as a modelling assumption applicable to the specific case of written hiring signals.

A growing body of research shows that generative AI substantially improves productivity across knowledge-intensive work more broadly. Dell'Acqua et al. (2026) examined the use of GPT-4 among 758 consultants completing realistic knowledge-work tasks in a professional-services firm. Across many of the tasks studied, AI assistance increased productivity and improved the quality of completed work. This suggests that these productivity gains are not confined to a single occupation or task type, but extend across a broader range of knowledge-intensive activities.

This broader pattern also holds, and is documented in more direct detail, for writing-intensive tasks specifically. One of the earliest large-scale randomized experiments on this question was conducted by Noy and Zhang (2023). The authors studied 444 professionals performing occupation-specific writing tasks. They found that participants with access to ChatGPT reduced the time spent on a task by roughly forty percent while increasing quality ratings by around eighteen percent. Together with the findings of Dell'Acqua et al. (2026), this suggests that generative AI enables users to produce better written output with less effort. This mechanism is not unique to writing tasks in isolation. Rather, it reflects a broader relationship between generative AI and productivity in knowledge-intensive work, where producing high-quality written output forms an important component of task performance.

For the purpose of this thesis, the key contribution lies in the mechanism these productivity gains reveal. Across different settings, generative AI consistently reduces the effort required to produce polished written output while often improving its quality. In other words, AI changes the production process by lowering the cost of generating high-quality written communication. Because written application materials are themselves forms of written production, there are strong theoretical reasons to expect that this mechanism also applies to CVs, cover letters, and other written hiring signals.

This mechanism provides the conceptual bridge to the signaling model developed later in this thesis. The model builds on the literature reviewed above by treating AI as a technology that lowers the cost of producing a high-quality written signal. Whether this cost reduction is distributed uniformly across candidates is a separate question. The next subsection reviews the literature addressing this issue and discusses why generative AI may benefit different candidate types to different degrees.

2.4 AI, substitute or augmentor

The previous section established that generative AI reduces the cost of producing high-quality written work. However, an important remaining question is whether these productivity gains are distributed equally across individuals. To understand why they are not, this section first turns to the broader economics literature on technological change, before returning to the empirical evidence on generative AI specifically.

Acemoglu and Autor's (2011) task-based framework provides a starting point for understanding why new technologies affect workers unequally. They argue that technological change affects workers by altering how tasks are allocated between humans and machines. New technologies can either substitute for human capabilities by performing tasks previously carried out by workers, or complement human capabilities by increasing the productivity of tasks that continue to require human expertise. Whether a technology ultimately narrows or widens performance differences therefore depends on the relationship between the technology and the tasks individuals perform.

This task-based logic finds direct support in the empirical literature on generative AI. In their randomized experiment, Noy and Zhang (2023) found that participants with lower initial writing ability experienced larger improvements in both writing quality and task completion time than stronger writers. They argue that generative AI substitutes for capabilities that less-skilled individuals previously had to supply themselves, allowing these users to narrow the performance gap with stronger writers. A similar pattern is observed outside writing-specific settings. Studying thousands of customer-support agents using a generative AI assistant, Brynjolfsson et al. (2023) find that less experienced and lower-performing workers benefited substantially more than their higher-performing colleagues. The technology increased productivity across the workforce, but the largest improvements were concentrated among those who initially performed below average. The consistency of this finding across different occupations suggests that heterogeneous productivity effects are not unique to writing tasks, but reflect exactly the kind of task-dependent substitution that Acemoglu and Autor's framework predicts.

This compression effect is not unconditional, however. Dell'Acqua et al. (2026), studying 758 consultants at a professional-services firm using GPT-4 across a range of realistic knowledge-work tasks, describe a "jagged technological frontier" separating tasks that AI can reliably support from those it cannot. Within this frontier, weaker performers again benefit disproportionately from AI assistance. Outside the frontier, tasks require complex judgment, open-ended reasoning, or genuinely novel problem solving. Consequently, when tasks fall outside the frontier, AI assistance may reduce the performance of weaker workers who struggle to recognize when a task exceeds the technology's capabilities.

Building on this task-based logic, Ide and Talamàs (2025, 2026) formalize the distinction between AI as an equalizer and AI as an augmentor, providing a theoretical account of the compression and divergence patterns documented above. When AI primarily substitutes for capabilities that lower-skilled individuals previously lacked, it compresses differences between workers by disproportionately improving the performance of initially weaker individuals. Conversely, when AI primarily complements capabilities already possessed by highly skilled individuals, it reinforces existing differences because those individuals are better able to use the technology effectively. This distinction provides a theoretical explanation for why empirical studies sometimes find larger benefits for weaker performers, while other contexts suggest the opposite.

The implications of these mechanisms for hiring are discussed by Lievens and Dunlop (2025). They argue that generative AI can play different roles in personnel selection depending on how applicants use the technology and on the nature of the assessment itself. AI may function as a substitute, replacing capabilities that applicants would otherwise need to demonstrate independently. It may act as an equalizer by allowing lower-quality candidates to produce application materials that more closely resemble those of stronger candidates. Alternatively, it may function as an augmentor when applicants with stronger underlying abilities are better able to evaluate, adapt, and improve AI-generated output. Rather than assuming that generative AI uniformly weakens or strengthens written application materials as hiring signals, this perspective suggests that its effect depends on which of these mechanisms dominates.

This distinction forms the theoretical foundation for the signaling model developed in the following sections. The model therefore does not assume in advance that generative AI either erodes or preserves the credibility of written application materials. Instead, it examines how different distributions of AI's cost-reducing effects across candidate types influence the conditions under which written application materials continue to function as credible signals of underlying ability.

3 Theory: From Literature to Model

The literature reviewed above establishes two premises that this thesis takes as its starting point. First, hiring is characterized by information asymmetry: the employer cannot directly observe a candidate's underlying quality and must infer it from observable, costly signals (Akerlof, 1970; Spence, 1973). Second, written application materials such as CVs, cover letters, and the general polish of written communication function as such signals. This is because producing a well-written application is assumed to be less costly for a high-quality candidate than for a low-quality one (Connelly et al., 2011). The credibility of this signal, in the traditional Spence framework, does not depend on the written application being difficult to produce in an absolute sense. Instead, the credibility of a written signal rests on a written application being differentially costly to produce across candidate types. It is this differential-cost property that allows the employer to infer the candidate type from the written signal.

This is precisely the property that generative AI puts under pressure. The productivity literature summarized in Section 2.3 shows that generative AI functions as a shock to the cost of producing written materials, and that this shock is not uniformly distributed across users. Noy and Zhang (2023) and Brynjolfsson et al. (2023) both find that initially weaker performers gain more from AI assistance than initially stronger performers. Dell'Acqua et al. (2026) refined this pattern by introducing the concept of a "jagged technological frontier," which explains that AI's benefits vary across tasks depending on how well they align with the technology's current capabilities. Within the frontier of AI capability, the same compression of the performance gap between weaker and stronger workers appears. However, outside the frontier, AI assistance may reduce the performance of weaker workers who struggle to recognize when a task exceeds the technology's capabilities. Taken together, this literature motivates treating AI as a parameter that changes the cost of producing a written signal, with this cost changing non-uniformly across candidate types.

3.1 Two competing mechanisms: equalizing versus augmenting

Building on the empirical findings of Noy and Zhang (2023), Brynjolfsson et al. (2023), and Dell'Acqua et al. (2026), this thesis formulates several assumptions that serve as a foundation for the modelling framework. These studies examine how the benefits of generative AI are distributed among different types of users. Building on their insights, this section explores the possible ways in which generative AI can function as a parameter for cost reduction in producing written signals within the hiring process. Exploring this within the hiring context involves an assumption that extends beyond the scope of the original studies. Connecting back to the literature, two competing mechanisms explain the differences in how these benefits are distributed across candidate types. As reviewed in Section 2.4, generative AI can operate either as an equalizer or as an augmentor of existing ability.

To further advance the debate on generative AI as an equalizing versus augmenting mechanism, Section 4 translates the theoretical discussion into a formal model. In order to bridge this translation, the modelling choices are motivated in the present Section. The first and most basic premise is that access to generative AI lowers the cost of producing a high-quality written application for all. Candidates who use AI assistance can reach a given quality in less time or effort than candidates who do not. This premise, on its own, says nothing about which candidate type benefits more. It treats the cost reduction simply as a uniform shock that applies to every candidate in the same way. Taken in isolation, this is the assumption behind the first scenario developed in the model below: AI as a symmetric cost reduction.

The second premise refines this by taking into account the heterogeneity documented directly in Noy and Zhang (2023) and Brynjolfsson et al. (2023), in which the weaker applicant benefits more from generative AI assistance than a stronger applicant. Dell'Acqua et al. (2026) locate this compression specifically inside the "jagged technological frontier." Assuming this compression, AI is represented as a reduction in the low type's cost of producing the written signal compared to that of the high type. Take into consideration, for example, the creation of a polished CV. In the pre-AI setting, the high type would produce a polished CV more quickly and with higher quality, whereas the low type can now generate a comparable CV at a more similar pace. Correspondingly, the second scenario is modeled as AI as an equalizer.

The third premise builds on the asymmetry inherent in Dell'Acqua et al.'s (2026) jagged frontier, discussed in Section 2.4. Outside the frontier, AI assistance can lower rather than raise performance. Weaker performers are possibly less able to recognize when a task exceeds the technology's capabilities. This thesis interprets this asymmetry as evidence that AI does not reduce the cost of producing a high-quality signal uniformly across candidate types. Where judgment is required to identify and correct the limitations of AI-generated output, this judgment itself constitutes a form of underlying ability. Candidates who possess such judgment are better positioned to convert AI assistance into high-quality material at lower cost. Formally, this translates into a cost reduction that is larger for the high type than for the low type. This motivates the model's third scenario, the opposite of the equalizer case considered above. In this scenario, AI functions as an augmentor of existing ability rather than a substitute for it.

3.2 The focus of the model

The signaling framework developed in the previous sections can be formalized in several ways. Four candidate variables could serve as the focal parameter: cost, the wage premium Δ, quantity, and quality. Each choice would emphasize a different mechanism and lead to distinct theoretical implications. This thesis treats cost as the focal parameter, while quantity, quality, and the wage premium Δ are fixed. The remainder of this section explains why.

Quantity can be dismissed. Nothing in the reviewed literature suggests that the number of signals produced materially affects credibility. Quantity is more relevant to questions of applicant-pool congestion, which fall outside the scope of this thesis.

Quality has received more direct attention in the literature and could in principle serve as the focal variable. In this thesis, however, quality is not treated as the mechanism itself. The candidate's underlying signal choice is defined as $q \in \{0, 1\}$. Here, $q = 1$ denotes a high-quality written application, and $q = 0$ denotes a low-quality one. Within the signaling framework, the causal sequence runs as follows. Candidates first choose the signal q. They then incur the cost associated with that choice. The signal choice therefore reflects the underlying cost structure. It does not itself drive credibility. This interpretation is consistent with the single-crossing property underlying the signaling framework in Section 2.1. Credibility rests on cost differences between types, not on the signal choice in isolation. Accordingly, the model treats q as the candidate's binary signal choice.

A similar argument applies to the wage premium Δ. From the employer's perspective, Δ is important because it captures the premium at which each type may choose to signal. Yet focusing on Δ would shift the analysis away from applicant behavior toward wage-setting decisions, which fall outside the scope of this thesis. The wage premium therefore remains in the model as a fixed parameter.

Cost is introduced as the model's focal parameter. Each candidate type i is characterized by a type-specific parameter θ_i. The baseline cost of producing the written signal in the pre-AI era, following Spence (1973), is formulated as $\frac{1}{\theta_i}$. Generative AI is modelled as α_i, the AI-induced proportional reduction in signaling cost, which scales this baseline cost downward. Together, θ_i and α_i determine the effective cost each type faces in producing a given signal choice q.

Cost is chosen as the focal parameter to maintain conceptual consistency with Spence's (1973) signaling framework, in which differences in signaling costs across types underpin credible signal formation. The credibility of a written signal does not depend on cost in an absolute sense, but on the single-crossing property. It is this differential-cost property, and not the wage premium Δ or the signal choice q themselves, that sustains a separating equilibrium in which employers can infer candidate type from observed behavior. Holding Δ and q fixed follows directly from Spence's logic, in which cost asymmetry is the structural mechanism generating credible signaling in the first place. Treating cost as the focal parameter allows the model to isolate the specific mechanism through which generative AI may disrupt or preserve the separating equilibrium.

4 Model

To examine how generative AI affects the credibility of written signals in hiring, a baseline model is constructed. The baseline model draws on the signaling framework of Spence (1973). Furthermore, the model is structured following the approach of Coles et al. (2013), who started with a simple formal setup followed by equilibrium characterization. The model presented here is kept simple on purpose. The model contains two candidate types, a binary signal and a cost function that captures the single-crossing property. The purpose of Section 4.1 is to establish the conditions under which written signals are informative before generative AI is introduced. This setup allows the effects of introducing AI, discussed in Sections 4.2–4.6, to be evaluated against a clearly defined baseline.

4.1 Pre-AI baseline

Consider two candidate types, $i \in \{H, L\}$. H denotes the high-type candidate and L the low-type candidate. The productivity of H is assumed to be higher than that of L expressed in y_i, $y_H > y_L$.

The employer is not able to directly observe the productivity of the applicant. The employer can instead only observe a written signal, $q \in \{0, 1\}$. The notation q is binary, where $q = 1$ denotes a high-quality written application and $q = 0$ denotes a low one. Producing the high-quality signal is costly. Following Spence (1973), the cost is assumed to differ systematically across types: $c_i(0) = 0$, $c_i(1) = \frac{1}{\theta_i}$ with $\theta_H > \theta_L > 0$. Given, $\theta_H > \theta_L$ it follows that $\frac{1}{\theta_H} < \frac{1}{\theta_L}$. The high-type candidate can produce the high-quality written signal at a strictly lower cost than the low-type candidate. This is the single-crossing property on which the credibility of the signal is based.

The labor market is assumed to be perfectly competitive. Employers earn zero expected profit in equilibrium, and wages equal a candidate's expected productivity given employers' beliefs about type. Under separating beliefs, where $q = 1$ is believed to come from H and $q = 0$ from L, competitive wage-setting implies: $w(q = 1) = y_H$, $w(q = 0) = y_L$. The candidate's decision to produce a high-quality signal can be formalized by the utility function. The utility function can be formalized as the following: $U_i(q) = w(q) - c_i(q)$, where $w(q)$ is the wage conditional on the employer's belief and $c_i(q)$ is the type-specific cost of producing the signal. The value of being recognized as a high-type candidate is captured by the wage premium $\Delta = y_H - y_L \succ 0$. The wage premium is the employer's belief-dependent component of $U_i(q)$ that shapes incentives. The candidate's decision depends on how this premium compares to the cost differences across types. When the premium sufficiently exceeds these costs, a separating equilibrium exists.

Moving forward, the separating equilibrium takes on the following form: $H \rightarrow q = 1,\ L \rightarrow q = 0$. The separating equilibrium is supported by two incentive-compatibility constraints.

The high-type candidate must prefer to send the costly signal, resulting in the following incentive-compatibility constraint for the high-type, IC-*H*:

$$U_H(1) \geq U_H(0) \Rightarrow w(1) - c_H(1) \geq w(0) - c_H(0) \Rightarrow y_H - \frac{1}{\theta_H} \geq y_L - 0 \Rightarrow \Delta \geq \frac{1}{\theta_H}$$

The low-type candidate must prefer not to imitate, formulated as IC-*L*:

$$U_L(0) \geq U_L(1) \Rightarrow w(0) - c_L(0) \geq w(1) - c_L(1) \Rightarrow y_L - 0 \geq y_H - \frac{1}{\theta_L} \Rightarrow \Delta \leq \frac{1}{\theta_L}$$

Combining IC-*H* and IC-*L* results in the baseline separating condition: $\frac{1}{\theta_H} \leq \Delta \leq \frac{1}{\theta_L}$. This forms the baseline result in the pre-AI situation. The separating condition implies that the required wage premium depends solely on the cost differential between the two candidate types in making the signal choice. The wage premium offered should be high enough to make it worthwhile for the high-type applicant to make the effort to produce the quality signal, but not so large that it becomes worthwhile for the low-type to imitate. Given $\theta_H \succ \theta_L$, this interval is always non-empty in the baseline, with width $W_0 = \frac{1}{\theta_L} - \frac{1}{\theta_H} \succ 0$. With no influence of AI, there is always some range of Δ for which the written signal separates types. This is the benchmark against which every AI scenario below is compared.

4.2 Generative AI as cost-reduction parameter

Generative AI is introduced as a parameter that reduces the cost of producing the high-quality written signal. No prior assumption is made regarding which type benefits more. Generative AI is formalized as a parameter; $\alpha_i \in (0, 1),\ i \in \{H, L\}$. This parameter captures the degree to which AI reduces the candidate's cost of producing the high-quality written signal q =1. An α_i close to zero means AI provides little assistance. An α_i close to one means AI makes the high-quality signal almost costless. The cost function after introducing AI is formalized as: $c_i(0) = 0,\ c_i(1) = \frac{1-\alpha_i}{\theta_i}$.

No relationship between α_H and α_L is imposed at this stage. Whether $\alpha_H \gtrless \alpha_L$ will later be modelled in line with the theory posed in Section 3. The modelling will formalize the distinction between the equalizer and augmentor functions of generative AI. Substituting the AI-modified cost function into IC-*H* and IC-*L*, these take the following form: with IC-*H*: $\Delta \geq \frac{1-\alpha_H}{\theta_H}$, and IC-*L*: $\Delta \leq \frac{1-\alpha_L}{\theta_L}$. By combining the AI-adjusted incentive compatibility constraints, the AI-modified separating condition is derived: $\frac{1-\alpha_H}{\theta_H} \leq \Delta \leq \frac{1-\alpha_L}{\theta_L}$. This separation condition is the central object of the model. A separating equilibrium exists only if the interval defined by this expression is non-empty.

Once the separating interval is established, its width relative to the pre-AI baseline indicates whether AI strengthens or weakens separation. The relationship is formalized through the expression $W_{AI} = \frac{1-\alpha_L}{\theta_L} - \frac{1-\alpha_H}{\theta_H}$, derived from the baseline function W_0 adjusted for the introduction of AI. The change in width relative to the baseline W_0 can be expressed as $W_{AI} - W_0 = \frac{\alpha_H}{\theta_H} - \frac{\alpha_L}{\theta_L}$. This expression is the criterion used throughout the remaining scenarios. The interval widens relative to the baseline exactly when $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$, and narrows when the reverse holds. Note that this is not the same as simply comparing α_H to α_L. Given $\theta_H > \theta_L$, the high type's cost reduction is divided by a larger denominator, so $\alpha_H > \alpha_L$ does not by itself guarantee $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$.

Every scenario evaluates whether, given a specific assumption about α_H and α_L, the interval widens, shrinks, or vanishes. It then examines how this outcome affects whether a polished written application remains a credible signal of candidate quality. This will be answered by clearly stating what is assumed in the given situation, followed by its implication for the separating equilibrium. Following this, the incentive-compatibility constraints will be assessed, and a real-world interpretation will be made.

4.3 AI as symmetric cost reduction

Building on the first premise posed in Section 3.2, AI is treated as a symmetric cost reduction, assumed to have $\alpha_H = \alpha_L = \alpha$, with $\alpha \in (0, 1)$. AI reduces both types' costs by the same proportional amount. When substituting the assumption into the separating equilibrium, this becomes: $\frac{1-\alpha_H}{\theta_H} \leq \Delta \leq \frac{1-\alpha_L}{\theta_L}$.

Both thresholds fall by the same factor $(1 - \alpha)$ relative to the baseline. The lower bound falls, which makes IC-*H* easier to satisfy. This implies that the high type needs a smaller wage premium to be incentivized to signal than before. But the upper bound also falls by the same factor, which makes IC-*L* harder to satisfy. This happens because the cost of imitation, where the low type must be deterred by, has shrunk. Applying the width criterion from Section 4.2, $W_{AI} - W_0 = \alpha\left(\frac{1}{\theta_H} - \frac{1}{\theta_L}\right) < 0$. For any $\alpha > 0$ the interval strictly shrinks relative to the baseline.

When we consider the real-world hiring context, it is not possible to simply state that symmetric AI assistance automatically completely destroys the credibility of the written signal. The high type may still retain a cost advantage in relative terms. What symmetric AI cost reduction has for implications is that it narrows the range of wage premiums for which separation can be sustained. As AI becomes more capable and α rises, fewer and fewer real-world wage premiums will fall inside the shrinking interval.

4.4 AI as equalizer

AI as an equalizer is assumed to benefit the low-type more than the high-type: $\alpha_L > \alpha_H$. This is the formal notation of the equalizer role of generative AI, as previously explained throughout Section 3.2 and is motivated by the findings in Noy and Zhang (2023) and Brynjolfsson et al. (2023). Substituting the assumption into the separation equilibrium constraint results in: $\frac{1-\alpha_H}{\theta_H} \leq \Delta \leq \frac{1-\alpha_L}{\theta_L}$

Applying the width criterion from Section 4.2 gives $W_{AI} - W_0 = \frac{\alpha_H}{\theta_H} - \frac{\alpha_L}{\theta_L}$. Because $\alpha_L > \alpha_H$ and $\theta_L < \theta_H$, both the numerator and denominator move in the same direction. This results in $\frac{\alpha_L}{\theta_L}$ being strictly larger than $\frac{\alpha_H}{\theta_H}$, so $W_{AI} - W_0 < 0$ always holds under the equalizer assumption. The interval strictly shrinks relative to the baseline, regardless of the specific magnitude of θ_H and θ_L.

IC-*H* is now easier to satisfy, since the high type's threshold falls. IC-*L* becomes harder to satisfy, since the low type benefits from a higher α than the high type. This larger effective cost reduction lowers the threshold for imitation, making it easier for the low type to mimic the high-quality signal. As α_L rises relative to α_H, the interval for Δ shrinks and can vanish entirely. This is the case when $\frac{1-\alpha_H}{\theta_H} > \frac{1-\alpha_L}{\theta_L}$, at which point no wage premium supports separation.

Once the interval vanishes, the written application can no longer support the separating equilibrium characterized in Section 4.1. In the model, the incentive compatibility constraints collapse, removing any effective distinction between candidate types. This thesis does not formally derive employer beliefs, hiring behavior, or a complete pooling equilibrium. It therefore refrains from asserting that employers necessarily hire more low-quality candidates as a consequence. The model's supported conclusion is limited to the finding that the separating equilibrium cannot be maintained.

4.5 AI as augmentor

AI as an augmentor is the opposing case to the equalizer scenario. Here, the assumption is that $\alpha_H > \alpha_L$. AI reduces the high type's cost of producing the high-quality signal by more than it reduces the low type's cost. This assumption is based on the high type having a better underlying ability to work with the technology than the low type (Dell'Acqua et al., 2026). Substituting the assumption into the separating equilibrium yields the expression $\frac{1-\alpha_H}{\theta_H} \leq \Delta \leq \frac{1-\alpha_L}{\theta_L}$.

$\alpha_H > \alpha_L$ alone does not guarantee that the lower bound falls by more than the upper bound. Applying the width criterion from Section 4.2 gives $W_{AI} - W_0 = \frac{\alpha_H}{\theta_H} - \frac{\alpha_L}{\theta_L}$. Given $\theta_H > \theta_L$, the high type's cost reduction α_H is divided by a larger denominator than the low type's. The interval widens relative to the baseline when $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$. The high type's augmentation advantage must be sufficiently large to offset the fact that AI operates on an already lower baseline cost. If α_H exceeds α_L modestly, this condition might fail even though $\alpha_H > \alpha_L$. The interval may narrow relative to the baseline, but the contraction is smaller than under an equalizer of equal magnitude.

IC-*H* becomes easier to satisfy; the high type's cost of producing a strong written application falls relative to its baseline cost. IC-*L* still tightens relative to the AI-free baseline, because the low type's cost of imitation also falls. However, it tightens less than in the equalizer scenario, where the low type receives the larger cost reduction. The augmentor case does not have a universal effect on interval width. Depending on the relative magnitudes of α_H and α_L, the interval under augmentation can shrink, remain unchanged, or expand relative to the baseline. The interval expands when $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$; it does not widen merely because α_H is larger in absolute terms.

In hiring markets, this scenario shows that when AI complements rather than substitutes for underlying ability, it may enhance the credibility of the written signal. This effect depends on the low type's incentive to imitate remaining limited. In practice, employers may benefit from applicants' use of AI. A broader range of wage premiums may remain consistent with separation, allowing candidate types to stay distinguishable.

4.6 Boundary case: near-costless AI

As a short limiting case rather than a full scenario, consider $\alpha_H \to 1$ and $\alpha_L \to 1$ simultaneously. At this limit, generative AI becomes sufficiently powerful to produce a polished written application at virtually zero cost for both types. Both cost thresholds, $\frac{1-\alpha_H}{\theta_H}$ and $\frac{1-\alpha_L}{\theta_L}$ converge to zero, and the separating interval collapses. No positive wage premium can satisfy this condition. As AI approaches this level of capability, the distinction between its role as an equalizer or augmentor becomes immaterial. Once the written signal becomes nearly costless for every candidate, it loses the differential-cost property on which its credibility depends. Therefore, the written signal could cease to function as a meaningful hiring signal.

4.7 Summary

Table 1 summarizes the four cases developed above.

Scenario	**Assumption**	**Main effect on costs**	**Implication for separation**
Baseline (A)	$\alpha_H = \alpha_L = 0$	Standard costly signaling; ordering $\theta_H > \theta_L$ unaffected	Separation possible whenever $\frac{1}{\theta_H} \leq \Delta \leq \frac{1}{\theta_L}$
Symmetric AI (B)	$\alpha_H = \alpha_L = \alpha$	Both costs fall by the same factor $(1-\alpha)$	$W_{\mathrm{AI}} - W_0 < 0$: interval shrinks as α rises and vanishes as $\alpha \to 1$
Equalizer (C)	$\alpha_L > \alpha_H$	Low type's cost falls relatively more; IC-L tightens	$W_{\mathrm{AI}} - W_0 < 0$ always; interval shrinks and can vanish, so the separating equilibrium can no longer be supported
Augmentor (D)	$\alpha_H > \alpha_L$	High type's cost falls relatively more; IC-H relaxes	$W_{\mathrm{AI}} - W_0 \gtrless 0$ *only if* $\alpha_H/\theta_H \gtrless \alpha_L/\theta_L$; widening is conditional, not automatic

Table 1. AI cost-reduction scenarios and their effect on incentive compatibility and separation.

5 General discussion

5.1 Summary of formal findings

The model developed in Section 4 shows that generative AI does not have a determinate effect on the credibility of written application materials as hiring signals. Whether a separating equilibrium survives the introduction of AI depends on the relative size of the cost-reduction parameters α_H and α_L, and on how these compare once scaled by each type's baseline cost parameter, θ_H and θ_L. The four cases summarized in Table 1 capture the main possibilities within the cost-reduction framework developed here.

Under a symmetric cost reduction, the range of wage premiums supporting separation narrows monotonically as AI becomes more capable. As a result, separation becomes increasingly fragile, while remaining feasible for any $\alpha < 1$. In the equalizer case, where the low type's cost falls disproportionately $\alpha_L > \alpha_H$, the admissible interval strictly shrinks relative to the baseline. In the augmentor case, where the high type benefits more $\alpha_H > \alpha_L$, the interval does not necessarily widen: it expands relative to the baseline when $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$. Otherwise, it still narrows, though less severely than under an equalizer of comparable magnitude. In the boundary case where AI drives both types' costs toward zero, the separating interval collapses regardless of the pattern of cost reductions.

5.2 Theoretical contribution

The contribution of this thesis is to make explicit the single mechanism on which the credibility of written hiring signals under generative AI turns: how AI alters the cost of producing the signal. The thesis does not aim to take a position on whether generative AI strengthens or weakens written hiring signals, but to clarify the conditions under which each outcome arises. Robie et al. (2026) document employers' concern that generative AI enables low-ability candidates to imitate the polished output of high-ability candidates, thereby eroding the validity of written applications. The model shows that this concern is formally justified precisely in the equalizer case, where $\alpha_L > \alpha_H$. It is most acute in the near-costless boundary case, where the signal loses its differential-cost property entirely.

At the same time, the model demonstrates that AI does not necessarily erode signal validity. In the augmentor case, and specifically when $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$, the written signal can remain as informative as before AI, or become more informative. A central result of the model is that a larger absolute cost reduction for high-ability candidates does not by itself guarantee that the separating interval widens. The direction of the effect is determined by the ordering of $\frac{\alpha_H}{\theta_H}$ relative to $\frac{\alpha_L}{\theta_L}$, rather than by the absolute ordering of α_H and α_L. When the condition $\frac{\alpha_H}{\theta_H} > \frac{\alpha_L}{\theta_L}$ does not hold, the interval may narrow, even though $\alpha_H > \alpha_L$ may remain true.

This distinction is important because it shows that framing the debate as equalizer versus augmentor is overly simplistic. While augmentor-type AI adoption may be necessary to preserve or strengthen signal validity, it is not sufficient to determine the outcome entirely. The overall effect depends on how the cost reduction interacts with each type's productivity parameter. Both positions in the literature are therefore correct under different, precisely specified conditions, rather than one being generally right and the other generally wrong. This is the thesis's main theoretical contribution. It identifies the boundary condition, $\frac{\alpha_H}{\theta_H} \gtrless \frac{\alpha_L}{\theta_L}$, that separates the two positions. The analysis shows that the disagreement in the literature reflects a genuine ambiguity in how AI affects the cost of producing written signals across candidate types. Such disagreement cannot be resolved by empirical evidence alone without specifying the underlying mechanism.

This contribution is deliberately scoped. The model formalizes how generative AI changes the cost of producing a written signal and derives the conditions under which a separating equilibrium exists. It does not model employer beliefs off the equilibrium path, employers' hiring decisions, or a complete pooling equilibrium. Where a separating equilibrium fails to exist, the model supports only the conclusion that the written signal can no longer distinguish candidate types within its own terms. It does not make claims about what employers believe or how they would behave in that situation. This scope restriction isolates the cost-side mechanism cleanly. It also means that the practical implications discussed below should be read as consequences of that narrower formal result, not as direct predictions about employer behavior.

5.3 Implications for hiring and personnel selection

The model suggests that the informativeness of written application materials is not a fixed property of generative AI as a technology. Instead, it depends on which candidate type receives the greater cost reduction relative to its pre-AI cost of producing the signal. Tasks that reward polish and fluency as ends in themselves are more likely to fall within the "jagged frontier" described by Dell'Acqua et al. (2026), where AI performs reliably regardless of the user's underlying ability. In this setting, the model's equalizer case and the associated erosion of signal credibility are most plausible. Tasks that require judgment about when and how to deviate from AI-generated output are more likely to sit near the boundary of that frontier, where a candidate's own ability continues to shape the quality of the final output. This is the setting in which the model's augmentor case is more plausible.

This distinction offers personnel-selection researchers and practitioners a way to structure an otherwise unresolved empirical question: namely, which kind of task an application process is effectively testing, rather than treating "AI use in applications" as a single, undifferentiated phenomenon.

5.4 Practical implications for employers and candidates

For employers, the central practical implication is that AI-assisted written materials should not be treated as either uniformly credible or uniformly worthless. Because the model provides a theoretical account rather than direct empirical evidence about hiring outcomes, these implications should be read as considerations to weigh rather than as fixed recommendations. What the model suggests is that employers can usefully assess whether their application process still requires a capability that remains differentially costly across candidates, even with AI assistance available. That is, whether producing a strong response still depends on one's own judgment, reasoning, or expertise that AI cannot supply equally well to every applicant.

It can be doubtful whether producing a specific written application still requires a candidate's own judgment, reasoning, or expertise because of generative AI. In such cases, relying heavily on written materials alone risks moving the application process toward conditions resembling the model's equalizer case. This may justify greater reliance on complementary parts of the selection process. Examples include work samples, structured interviews, supervised or timed assessments, or tasks that specifically require judgment beyond easily generated polish. These may be possible ways of reintroducing a differential-cost property that a purely written application may no longer reliably provide.

For candidates, the model implies that the return to underlying ability has not disappeared, but has become more conditional on the specific task and context. In the augmentor scenario, an underlying advantage possibly still translates into a lower cost of producing a strong signal. This may be possible through effective prompting and judgment about when to deviate from AI output. In the equalizer scenario, and especially near the boundary case, this incentive weakens, since both types can produce a similarly polished application at low cost using AI alone.

5.5 Limitations

The model is kept deliberately simple, and this simplicity brings limitations. It works with two discrete candidate types and a binary signal, rather than a continuum of ability and a continuous measure of signal quality. Whether the qualitative conclusions above survive under more realistic, continuous distributions of ability remains an open question. The parameters θ_H, θ_L, α_H, and α_L are motivated by the literature reviewed in Section 2 rather than estimated from real-world data. As a result, the model speaks to the direction of AI's effect on separation, not to its magnitude in any specific hiring market.

A further limitation concerns scope rather than realism. This thesis focuses specifically on how generative AI changes the cost of producing a written signal and derives the conditions under which a separating equilibrium exists. It deliberately does not model employer beliefs once separation fails: employers' hiring decisions, or a full pooling equilibrium. Extending the model in those directions would require additional assumptions about off-equilibrium beliefs and employer behavior, which fall outside the scope of this thesis.

Consequently, statements in Sections 5.3 and 5.4 about hiring outcomes are necessarily cautious. They should be read as plausible implications of the cost-side mechanism rather than as results the model formally derives. In line with Bergh et al.'s (2014) methodological point, this thesis has not empirically verified whether a separating equilibrium actually holds in AI-affected hiring markets. The model identifies the conditions under which it should hold, but testing whether real hiring markets satisfy those conditions is left to future research.

The model is also static. It does not analyze repeated hiring rounds, nor does it allow employers to respond strategically to a changing signal environment. For example, the application process could be redesigned, including tasks that are harder to substitute with generative AI. The wage premium Δ, candidate type, and signal quality are all held fixed. Signal quality is treated as binary, even though the quality of a CV or cover letter is likely to be influenced by AI use in ways the current model cannot capture. Similarly, while the wage premium has not adjusted proportionally to changes in the cost of producing a written signal, it is not fixed in reality. Treating it as constant is therefore a simplification that should be noted explicitly.

5.6 Future research

These limitations point toward natural extensions of this thesis. One direction is empirical. Future work could estimate α_H and α_L directly, for example through experiments in the style of Noy and Zhang (2023) applied to hiring-relevant writing tasks. This would allow an assessment of which of the four scenarios in Table 1 best characterizes current hiring markets.

The second direction is dynamic. A dynamic model in which employers adjust their screening practices across repeated hiring rounds could capture how employers respond as the written signal loses credibility. Such a model could possibly show whether the market converges toward a new, AI-resistant signal rather than remaining indefinitely in a degraded separating equilibrium.

A third extension follows directly from the scope limitation discussed in Section 5.5. The model could be extended to formally derive employer beliefs and, potentially, a complete pooling equilibrium. Rather than stopping once separation fails, this approach would connect the cost-side mechanism developed in this thesis to explicit predictions about hiring outcomes. This would move the model beyond identifying when separation breaks down. It would also begin to explain how employers are likely to behave in those cases.

Finally, extending the model to a continuum of candidate types, with signal quality chosen endogenously rather than fixed, would relax several simplifying assumptions made in Section 3.2. This would allow for a richer analysis of how AI affects written signals.

5.7 Conclusion

Using a simple signaling model built on Spence's (1973) incentive-compatibility logic, this thesis shows that the effect of generative AI on the credibility of written application materials in hiring is conditional rather than uniform. Whether that credibility is preserved, weakened, or lost depends on how AI's cost-reducing effect compares across candidate types relative to their own pre-AI costs. The mere use of AI assistance is not decisive. This conclusion provides formal support for the nuanced view proposed by Lievens and Dunlop (2025). It also situates the concern raised by Robie et al. (2026) as one coherent and formally identifiable case within a broader set of possible outcomes, rather than as the sole consequence of introducing AI into the hiring process.

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